2.2 BIASES UNDER CONSIDERATION

In either of the above two formulations, preference models may overprioritize certain features, which can cause misalignment or unreliable evaluation. As shown in Table 1, we study five biases that are idiosyncratic of LM generations:

- **Length**: Preference models often favor longer responses, even when the added length doesn't contribute substantive information (Singhal et al., 2024; Dubois et al., 2024; Shen et al., 2023). This bias may stem from a training data heuristic, where length is a correlate of comprehensiveness. As a result, models may generate overly verbose responses.
- **Structure**: Preference models may disproportionately favor responses with bullet lists or numbered points over narrative prose, even when prose is more suitable (Li et al., 2024). This bias may stem from structured formats being overrepresented in responses preferred by annotators. The consequence is a potential overuse of listicles, leading to outputs that feel formulaic or fail to convey arguments that benefit from prose. Prior work has further shown that such format biases, including preferences for lists, emojis, and other stylistic markers, are pervasive in preference models and can be easily amplified during alignment (Zhang et al., 2025).
- **Jargon**: This refers to a preference for responses using specialized or domain-specific terminology, even when it is not necessary. Models might learn this if the presence of jargon in the training data is correlated with highly preferred responses, leading them to use it as a proxy for quality. Resultingly, models may generate responses that give a superficial impression of expertise without being more useful.
- **Sycophancy**: This involves the model agreeing with or validating the user's stated opinions and assumptions, rather than offering a neutral and objective response (Sharma et al., 2024; Perez et al., 2023). This behavior may arise from training data if human annotators preferred sycophantic responses more often. The downside of this bias is that models may reinforce a user's biases, fail to provide objective information, and appear less trustworthy.
- **Vagueness**: This bias is characterized by models favoring responses that make broad statements that cover multiple aspects superficially, rather than providing concrete information that specifically addresses the query (example in Appendix Table 2). This may stem from vague statements being less falsifiable, and thus less penalized in training data. Such vague outputs can lead to responses that are unhelpful and lack depth.

2.3 COUNTERFACTUAL TESTING

Why Counterfactual Data? Preference models may rely excessively on the above features. To measure their reliance on these features, simple correlation analysis is insufficient because it can conflate multiple features. We construct counterfactual response pairs that differ primarily in the expression of a target feature, while other features remain consistent (e.g., responses with *more* or *less* jargon with roughly equal lengths).

Creating Counterfactual Pairs For each query Q and base response R, we apply a perturbation function f_p to obtain a perturbed response $R'_p = f_p(R)$, where p is a bias feature. Ideally, the perturbation function f should only change the bias feature, not other core aspects of R that would impact how a preference model scores R'_p.

To approximate this perturbation function f_p, we use the RATE (Rewrite-based Attribute Treatment Estimators) protocol (Reber et al., 2025). Specifically, we first use a language model to rewrite an original base response to produce a counterfactual response R'_p that amplifies the bias feature p. We then correct for the rewriting error by rewriting again to produce a new base response R_p. Using the pair (R_p, R'_p), we compute $W_{RM}(Q, R_p)$ and $W_{RM}(Q, R'_p)$ to measure the causal effect of p on the reward score. Similarly, we compute $W_{\text{EVAL}}(Q, R_p, R'_p)$ to measure the effect of p on W_{EVAL}'s evaluation judgment.

Human Evaluation To evaluate overreliance on these features, we need to understand how humans perceive these counterfactual responses. Therefore, we collect human preference judgements for 100 randomly sampled (Q, R_p, R'_p) triples for each bias feature p. We collect 3 judgements per query, and compute the final judgement through majority voting. Additional details are in Appendix A.

Metrics For each bias feature p, we compute the below metrics:

- **Skew Rate**: This is the frequency with which the preference model W favors the perturbed response $R_p'^{(i)}$ over the base response $R_p^{(i)}$. Let the difference between the scores for the two responses be $\Delta s_i = W_{RM}(Q^{(i)}, R_p'^{(i)}) - W_{RM}(Q^{(i)}, R_p^{(i)})$.

$$\text{Skew}_p = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\Delta s_i > 0) \tag{4}$$

  Here, Skew_p is 1 if the model prefers the perturbed response $R_p'^{(i)}$ and 0 otherwise.

- **Miscalibration Rate**: This measures the degree of disagreement between the model's preference and the aggregated human majority preference for $R_p'^{(i)}$ over $R_p^{(i)}$ (denoted as $\text{Human}(R_p'^{(i)} > R_p^{(i)})$).

$$\text{Miscal}_p = \frac{1}{N}\sum_{i=1}^{N} |\mathbb{I}(\Delta s_i > 0) - \mathbb{I}(\text{Human}(R_p'^{(i)} > R_p^{(i)}))| \tag{5}$$

  Similarly, $\mathbb{I}(\text{Human}(R_p'^{(i)} > R_p^{(i)}))$ is 1 if the aggregated human majority vote favors $R_p'^{(i)}$ over $R(i)_p$ and 0 otherwise.

Miscalibration measures the degree to which a model's preferences align with human judgments and skew indicates the model's intrinsic preference rate for biased responses. While an ideal preference model would achieve zero miscalibration ($\text{Miscal}_p = 0$), its Skew_p (i.e., rate of favoring responses with the perturbed feature p) should ideally be close to the observed HumanSkew_p. This is because the HumanSkew_p indicates the baseline extent to which human evaluators find responses exhibiting the amplified feature p to be preferable.

3 ARE PREFERENCE MODELS MISALIGNED WITH HUMAN JUDGEMENTS?

Dataset. To study structure, jargon, and length biases, we sample 100 queries from Chatbot Arena (Chiang et al., 2024), where we filter single-sentence queries that are in English, grammatically well-formed (ending with a question mark), semantically meaningful and non-offensive. Since sycophancy is plausible when the user expresses an opinion, we generate 100 queries that express an opinion (e.g., *"Isn't modern art just lazy compared to classical techniques?"*). To study vagueness, we use 78 human-written NLP-related queries from the KIWI dataset (Xu et al., 2024), and generate 22 additional queries that cover similar content. Scientific queries inherently require unambiguous and detailed responses and a preference for vague responses would represent a clear flaw. For all these queries, we generate counterfactual pairs of responses using the RATE protocol. Prompts are provided in Appendix B.

Models. We consider four reward models, all trained on v0.2 of the Skywork reward data collection (Liu et al., 2024), which aggregates diverse preference datasets (e.g., HelpSteer2, OffsetBias, WildGuard) and underlies many top-performing open-source reward models. Specifically, we study Gemma2-2B, Gemma-2-27B, Llama-3.1-8B, and Llama-3.2-3B. In addition, we consider three proprietary models that we use as LLM evaluators: Gemini-2.5-Pro (Team et al., 2023), GPT-4o (Hurst et al., 2024), and Claude-3.7-Sonnet (Anthropic, 2025). For response generation, we used GPT-4o.

Results. As shown in Figure 2, our analysis of preference models (reward models in top row and LLM evaluators in bottom row) shows that these models consistently show miscalibration and a high rate of skew in favoring perturbed responses across various bias categories.

Reward Models. Reward models exhibit clear miscalibration relative to human judgments: model preference rates for perturbed responses systematically deviate from human preference rates. **While *vagueness* and *jargon* elicit the highest miscalibration (>50%), length and sycophancy also show substantial miscalibration.** This suggests that models struggle to align with human judgments when responses contain overly technical language or lack specificity. An example of this is shown

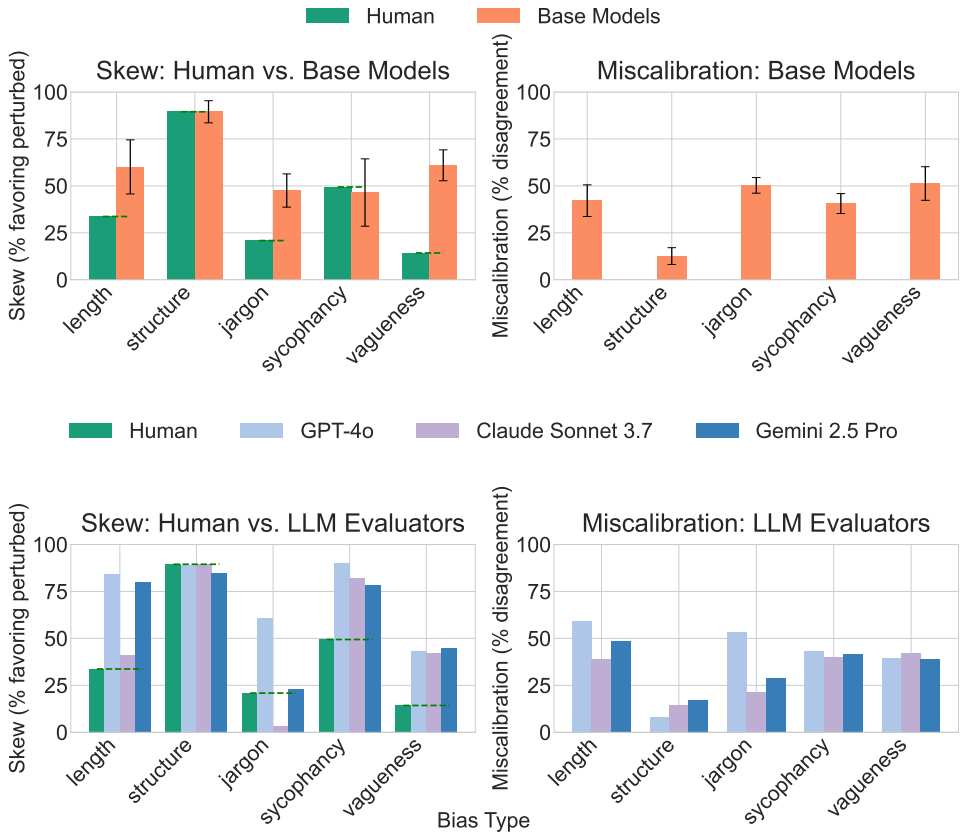

Figure 2: Skew and calibration errors averaged across reward models (top row) and all LLM evaluators (bottom row) in favor of perturbed (biased) responses, compared with human preferences.

for *vagueness* is shown in Table 2, where the model prefers a response that contains many broad claims that lack specificity. Interestingly, reward models exhibit the lowest miscalibration for the *structure* bias (∼15%), indicating that their preference for structured responses aligns more closely with human preferences. Similarly, models also show a higher preference for responses containing jargon and vague statements, with large differences in human and model skew rates. At the same time, the models' tendency to prefer agreeable responses mirrors human tendencies based on similar skew rates for sycophancy.

LLM Evaluators. Figure 2 shows that LLM evaluators also exhibit significant miscalibration relative to human preferences, with the largest deviations in *length* and *vagueness* biases. LLM evaluators similarly amplify skew toward perturbed responses compared to human annotators, particularly for *vagueness* and *sycophancy*. Notably, **LLM evaluators show a dramatically higher preference for sycophantic responses (∼75-85% skew) compared to humans (∼50%)**. These findings reveal that preferences of LLM evaluators can similarly diverge from human preferences.

4 Are preference models over-reliant on biases in training data?

To investigate whether the biases observed in preference models (§3) might originate from the training data, we first analyze the Skywork dataset, on which all four reward models were trained. Our analysis focuses on identifying skews in human preferences within this data and then quantifying how these skews correlate with model behavior.

Training data skew. Each bias is first mapped to a measurable quantity that can be automatically extracted from a response; for example, for *length* we record the token-count of the response, whereas for *structure* we use a binary flag that detects the presence (or absence) of list-style formatting. The exact labeling heuristics used for all biases are provided in Appendix C. For each query-response pair